Project Title

RGB-D microtopography of sandprawn bioturbation

Reference Number (Pre-Screening Questionnaire)

STU-EBE-2026-PSQ002548

Application committee

Faculty of Engineering & Built Environment

Student or Researcher application

Student

Abstract

This project develops an RGB-D microtopography pipeline to quantify sandprawn bioturbation, in collaboration
with the Department of Biological Sciences at UCT. The work is based on the framework of Azhar et al. (2022), but
adapted to the Intel RealSense camera. The main tasks are to calibrate the sensor, characterise its depth noise,
and develop a processing pipeline that extracts the detrended sediment microtopography and the roughness
metrics that describe surface disturbance. The system will be validated against controlled targets of known relief,
so that the depth accuracy and roughness measurements can be quantified. As an extension, the project will also
detect and segment the specific surface features produced by sandprawns, including mounds and burrow openings, using classical morphological image-processing methods. This allows the project to move beyond general
roughness metrics and measure the visible bioturbation features directly

M. Azhar, J. R. Hillman, T. Gee, S. Schenone, W. van der Mark, S. F. Thrush, and P. Delmas. An RGB-D framework
for capturing soft-sediment microtopography. Methods in Ecology and Evolution, 13(8):1730–1745, 2022. doi:
10.1111/2041-210X.13899

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